Project Report

DATA ANALYTICS WITH POWER BI

(Project Semester: August–December 2025)

CRASH REPORTING & DRIVER BEHAVIOR ANALYSIS

Submitted by

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Course Code: **INT374**

Under the Guidance of

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DECLARATION

I, **Ritik Raushan**, student of Computer Science & Engineering under CSE/IT discipline at Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 20/12/2025
Signature: Ritik Raushan
Registration No: 12300915

CERTIFICATE

This is to certify that **Ritik Raushan**, bearing Registration No. **12300915**, has completed the **Data Science Minor Project** titled **"Crash Reporting & Driver Behavior Analysis"** under my guidance and supervision. To the best of my knowledge, the present work is the result of his original development, effort, and study.

Signature of Supervisor
Ms. Sandeep Kaur
Designation: Assistant Professor
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Date: 20/12/2025

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my project guide **Ms. Sandeep Kaur** for her continuous support, guidance, and encouragement throughout the completion of this project. Her valuable insights and feedback greatly contributed to the successful completion of this work.

I also thank Lovely Professional University for providing the necessary resources and learning environment. Lastly, I am grateful to my friends and family for their constant motivation and support.

TABLE OF CONTENTS

1. Introduction
2. Scope of the Analysis
3. Existing System
    - Drawbacks of Existing System
4. Source of Dataset
5. Data Preprocessing / Cleaning
6. Data Modelling
7. Analysis on Dataset
8. List of Analysis with Results
9. Future Scope
10. References

1. INTRODUCTION

Road accidents are a major concern worldwide, causing loss of life, injuries, and economic damage. Analyzing crash data helps identify patterns related to driver behavior, road conditions, vehicle types, and risk factors.

This project focuses on building an **interactive Power BI dashboard** to analyze crash reporting data and driver behavior. The dashboard provides insights into crash trends, injury severity, driver fault, vehicle involvement, and risk prediction to support data-driven decision-making.

2. SCOPE OF THE ANALYSIS

The scope of this project includes:

- Analyzing crash trends over time
- Understanding driver behavior patterns
- Identifying high-risk vehicles and manufacturers
- Studying injury severity and crash causes

- Creating interactive dashboards for visualization
- Supporting safety-oriented decision-making

3. EXISTING SYSTEM

Traditional crash data analysis is mostly performed using static reports, spreadsheets, or manual methods. These systems lack real-time insights, interactivity, and advanced visualization.

Drawbacks of Existing System

- Static and non-interactive reports
- Time-consuming analysis
- Limited visualization capabilities
- Difficult to identify risk patterns quickly
- No predictive or trend-based insights

4. SOURCE OF DATASET

https://catalog.data.gov/dataset/crash-reporting-drivers-data

The dataset used for this project is a **Crash Reporting – Drivers Dataset**, containing information such as:

- Crash date and time
- Driver details
- Injury severity
- Vehicle make, model, and type
- Collision type
- Traffic control conditions
- Speed limits

5. DATA PREPROCESSING / CLEANING

The following preprocessing steps were applied:

- Removal of duplicate records
- Handling missing and null values
- Standardizing categorical values
- Creating calculated columns
- Data type correction
- Feature selection for analysis

These steps ensured data accuracy and consistency.

6. ANALYSIS ON DATASET

i. Introduction

Multiple analyses were performed to understand crash patterns and driver behavior.

ii. General Description

The analysis focuses on crash frequency, injury severity, vehicle involvement, and risk factors.

iii. Specific Requirements, Functions & Formulas

- Total Crashes (DAX Measure)
- Injury Count Measures
- Average Crashes per Period
- Top 10 High-Risk Vehicle Models
- Driver at Fault Percentage

iv. Analysis Results

- Passenger cars show the highest crash involvement
- Front-to-rear collisions are most frequent
- Majority of crashes result in non-fatal injuries
- Certain vehicle models are more accident-prone

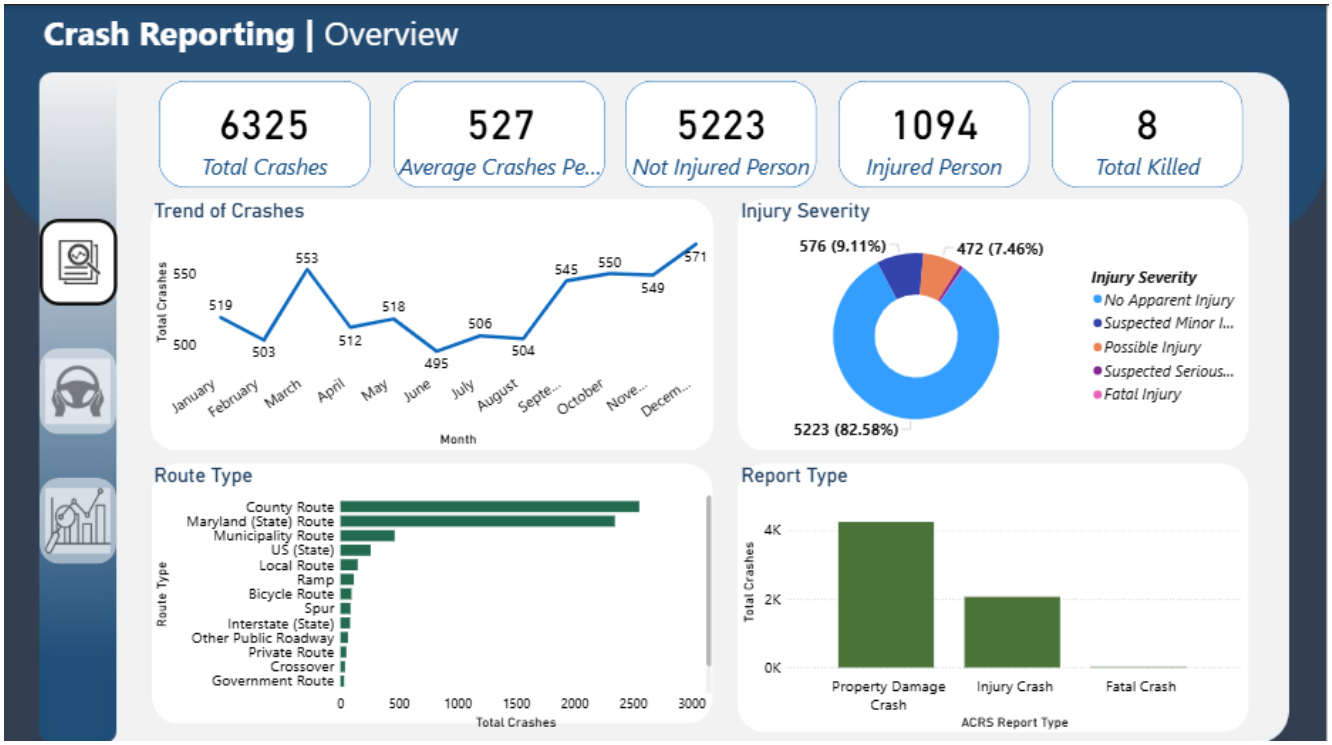
v. Visualization

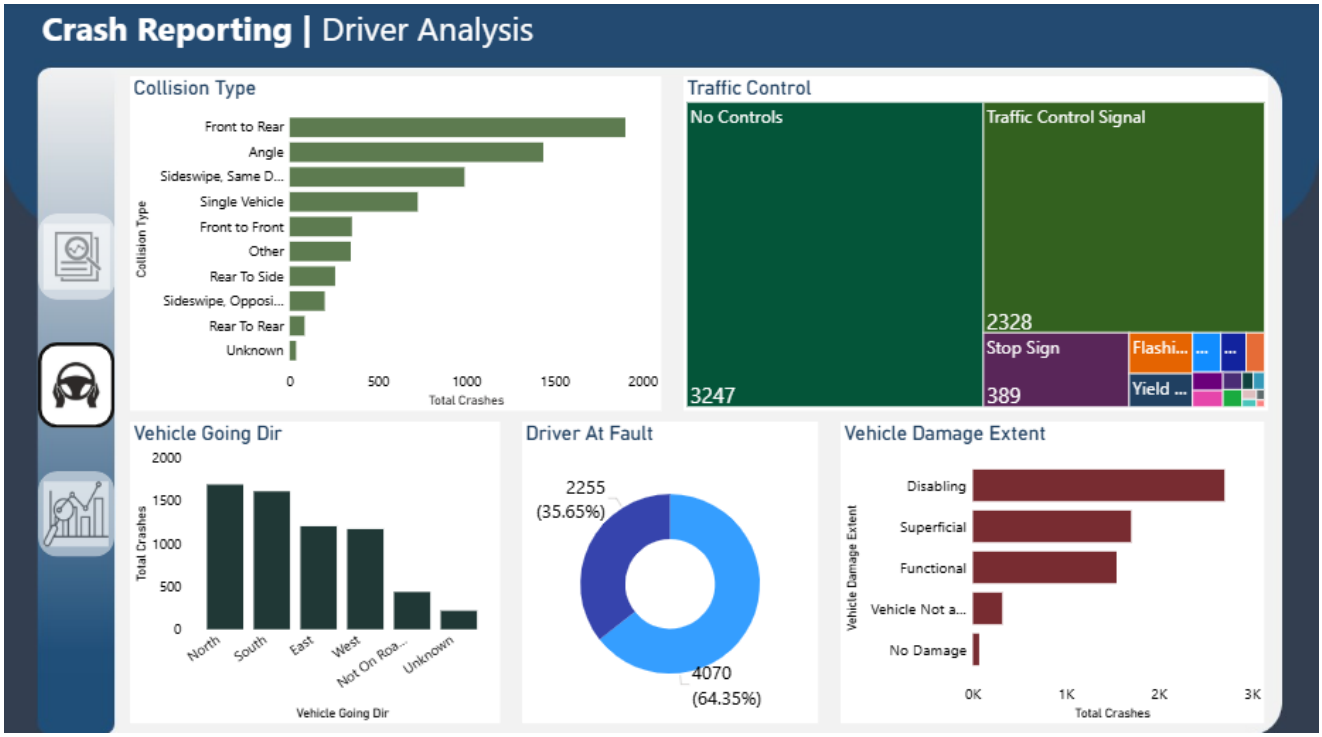
- KPI Cards
- Line Charts

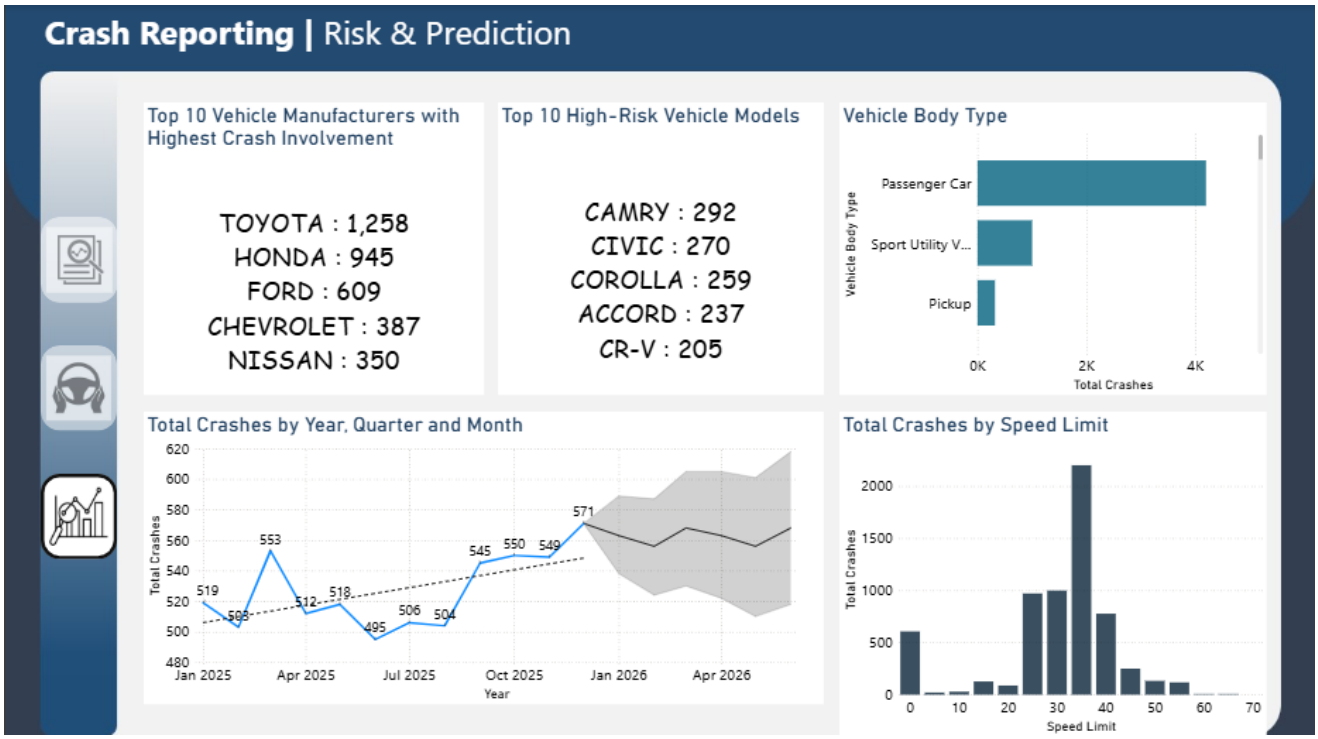
- Bar Charts
- Donut Charts
- Tree Map
- Predictive Trend Chart

7. LIST OF ANALYSIS WITH RESULTS

Analysis Type	Result
Crash Trend Analysis	Seasonal variations observed
Injury Severity	Majority non-fatal
Driver Fault Analysis	Drivers responsible in most cases
Vehicle Risk Analysis	Certain models show higher risk
Speed Limit Analysis	Higher crashes at mid-range speeds







8. FUTURE SCOPE

- Integration of real-time crash data

- Advanced ML models for prediction
- Accident hotspot analysis using GIS
- Driver risk scoring system
- Policy-level safety recommendations

9. REFERENCES

1. Microsoft Power BI Documentation
2. Road Safety and Crash Analysis Research Papers
3. IEEE Data Analytics Publications
4. Government Crash Reporting Portals

Dataset Link:

https://catalog.data.gov/dataset/crash-reporting-drivers-data

LinkedIn Link:

https://www.linkedin.com/posts/ritik314_powerbi-dataanalytics-crashanalysis-activity-7408110139045691392-ekqs?utm_source=share&utm_medium=member_desktop&rcm=ACoAAEUcgtkBBVgcAr4YfW0zKas6vIkNdRFAFkA

GitHub Link:

https://github.com/RitikRaushan314/PowerBI-Project